MOODY'S
INVESTORS SERVICE

SPECIAL COMMENT

Glossary of Moody's Ratings Performance Metrics

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Introduction

This Special Comment updates the original "Glossary of Moody's Ratings Performance Metrics," published in September 2011. It explains the metrics used in Moody's default, transition, and ratings performance studies. It is intended to serve primarily as a technical paper, explaining the metrics in sufficient detail to permit the reader to replicate the calculations. In a few cases it explains why calculations are done in certain ways rather than others (see most notably the discussion of withdrawal-adjusted versus unadjusted default rates), but mostly this publication details the facts of calculation. A broader discussion of how these metrics are used to measure ratings performance is included in Moody's Special Comment, "Measuring the Performance of Credit Ratings," published in November 2011.

The metrics are grouped into five broad categories:

1. Default Rates,[1] including default rates by cohort and original rating;
2. Loss Given Default (LGD) and Recovery Metrics;
3. Rating Transitions;
4. Ratings Volatility, including upgrade, downgrade and rating reversal rates; and
5. Ratings Accuracy, including metrics to measure relative or ordinal accuracy of a rating system such as cumulative accuracy profile, accuracy ratio, and average default position.

There is a supplemental glossary of frequently used terms following the detailed explanation of these metrics.

[1] All statistics calculated for defaults can also be calculated for impairments. We will ignore the distinction between defaults and impairments throughout this Special Comment.

Default Rate Metrics

Moody's calculates default rates in order to quantify and distinguish observed credit risk across rating categories for a portfolio of credits. Default rates are usually calculated by cohort rating but may occasionally be calculated by original rating. The nature of the calculations is the same in either case; the only difference is how credits are grouped together. In the "cohort" method, we sample ratings as of a particular date and group credits which have the same credit rating.[2, 3] In the "original rating" method, we group credits which have the same initial credit rating. The subsequent calculations are otherwise identical.

Cumulative Default Rate for a Single Set of Credits

We begin by calculating the default rate for a single set of credits. Let n_t denote the number of credits outstanding at the beginning of period t, let d_t denote the number of credits which default in period t, and let w_t denote the number of credits which withdraw in period t.[4] The initial set of credits is denoted n_0, and trivially we define $d_0 = 0$. Note that this initial set could be formed on the basis of having a common rating as of a cohort date, or by having a common initial rating, or by some other grouping criteria.

Moody's prefers to calculate *withdrawal-adjusted default rates*, meaning we treat rating withdrawals as censored observations in the Kaplan-Meier sense.[5] Under this method, we define the outstanding number of credits recursively as:

$$n_t = n_{t-1} - d_{t-1} - w_t \tag{1a}$$

The alternative, of course, is not to adjust for withdrawal. The resulting *unadjusted default rate* effectively assumes that withdrawn credits do not default. This requires the following modification to Equation (1a):

$$n_t = n_{t-1} - d_{t-1} \tag{1b}$$

The default hazard rate h_t is the probability a credit will default in period t given that it has not defaulted before period t. It is given by:

$$h_t = \frac{d_t}{n_t} \tag{2}$$

The cumulative default rate (CDR) for period t is the probability that a credit defaults some time before or during period t, and is calculated from default hazard rates. Let D_t denote the cumulative default rate, which is defined as:

$$D_t = 1 - \prod_{s=1}^{t}(1 - h_s) \tag{3}$$

Adjusted and Unadjusted Default Rates

The only difference between the adjusted and unadjusted default rates is whether one uses Equation (1a) or (1b) to determine the number of surviving credits over time. This immediately implies that the adjusted rate will never be less than, and almost always greater than, the unadjusted rate. Moody's prefers to report the adjusted rate because it answers what is generally a more interesting question. If one asks, "what is the probability that this credit will default within five years," the answer is the *unadjusted* default rate. But that question, and that answer, include the possibility that the credit will mature at some point within five years

2 "Credits" could refer either to issuers or issues.
3 Credits which are in default as of the cohort date are excluded from the cohort, even if they are rated.
4 Credits which are observed to default within one year of withdrawal are considered defaults, not withdrawals.
5 For further details please refer to Kaplan, E. L.; Meier, P. (1958), "Nonparametric estimation from incomplete observations." *J. Amer. Statist. Assn*. 53 (282): 457–481.

– mature, and thus not default. In the extreme, it might mature after the first day, and thus have essentially no possibility of defaulting within five years.

If instead one asks, “for a credit that will otherwise remain outstanding for five years, what is its probability of default,” the answer to that question is the *adjusted* default rate, and that seems to Moody’s to be the more interesting, relevant question. Faced with a credit scheduled to mature in one year, an investor would not sensibly ask about its five-year default risk. By implication then, when an investor does ask about five-year default risk, the presumption is that this applies to credits which are otherwise expected to remain outstanding for at least five years.

Of course there are exceptions where the unadjusted rate may be the appropriate measure. When considering a randomly formed static portfolio, an investor might want to know how many defaults she is likely to record within five years. If we are willing to assume that the maturity profile of this portfolio is “average,” then the unadjusted default rate is arguably the better answer. An even better approach would be to determine the exact maturity profile of the portfolio and apply the corresponding adjusted default rates to that maturity profile.

Average Cumulative Default Rate for Multiple Sets of Credits

The average cumulative default rate for multiple sets of credits is derived from the (weighted) average default hazard rates over the sets. Let $h_t(i)$ denote the hazard rate for the i^{th} set of credits for period t, exactly as defined above. Similarly let $n_t(i)$ and $d_t(i)$ denote the number of outstanding credits and the number of defaults, respectively, for the i^{th} set of credits for period t. Then the weighted average hazard rate taken over N sets for period t is given by:

$$\bar{h}_t = \frac{\sum_{i=1}^{N} n_t(i) \cdot h_t(i)}{\sum_{i=1}^{n} n_t(i)} = \frac{\sum_{i=1}^{N} d_t(i)}{\sum_{i=1}^{N} n_t(i)} \qquad (4)$$

The corresponding average cumulative default rate is analogous to Equation (3), and is given by:

$$\bar{D}_t = 1 - \prod_{s=1}^{t} \left(1 - \bar{h}_s\right) \qquad (5)$$

We provide the example below to illustrate the calculations for adjusted default hazard rates and cumulative default rates. The example considers two cohorts; the first cohort consists of 200 like-rated credits as of January 1, 2013, while the second cohort consist of 150 like-rated credits as of January 1, 2014.

An Example of Calculating Cumulative Default Rates

2013			2014		
1/1/2013	12/31/2013		1/1/2014	12/31/2014	
Number Outstanding	Defaulted	Withdrawn	Number Outstanding	Defaulted	Withdrawn
200	10	95	95 (from 2013)	5	25
			55 (new ratings)	10	20

In this example, the first-year default hazard rate for the January 1, 2013 cohort is $h_1(1)$ = 10 / (200 - 95) = 9.52% and the second-year default hazard rate is $h_2(1)$ = 5 / (95 - 25) = 7.14%. Therefore, the two-year adjusted CDR for this cohort is $D_2(1)$ = 1 - (1 - 9.52%)*(1 - 7.14%) = 15.99%. Note that the two-year unadjusted CDR is simply 15 / 200 = 7.50%, less than half the default risk implied for credits which would have otherwise lasted two years.

The first-year default hazard rate for the January 1, 2014 cohort is given by $h_1(2)$ = (5 + 10) / (95 + 55 - 25 - 20) = 15 / (150 - 45) = 14.29%. The average first-year default hazard rate is thus h_1 = (10 + 15) / (200 - 95 + 150 - 45), or 11.90%. Since a second-year default hazard rate can only be calculated for the January 1, 2013 cohort, the average second-year default hazard rate is trivially 7.14%. Therefore, the average two-year CDR is D_2 = 1 - (1 - 11.90%)*(1 - 7.14%) = 18.20%.

As discussed above, we calculate default rates by cohort and original rating using identical methods. The only difference is that while the cohort method groups credits by their rating on a particular calendar date and tracks their performance by calendar time, the original rating method groups credits by their original rating and tracks their performance by their elapsed time (seasoning).

For example, the one-year single-B default rate for the January 1, 2014 *cohort* refers to (non-defaulted) credits rated single-B on that date, irrespective of when or at what rating they were first issued, and tracks which of those defaulted in 2014. In the one-year default rate by the original rating method, the credits are grouped by their original rating and the numerator refers to credits that defaulted during the first 12 months after that assignment without regard to calendar time. The following example illustrates the difference between the two methods.

An Example Illustrating the Difference between Default Rates by Cohort Rating and Original Rating

2013			2014		
1/1/2013	12/31/2013		1/1/2014	12/31/2014	
Number Issued	Defaulted	Withdrawn	Number Outstanding	Defaulted	Withdrawn
100, rated Baa	0	0	95, remain Baa	0	45
			5, downgraded to single-B	5	0
100, rated single-B	0	0	100, remain single-B	10	35

In this example, 100 Baa-rated and 100 single-B rated credits are issued on January 1, 2013. By the end of 2013, 95 of the 100 Baa-rated credits remained Baa and five credits were downgraded to single-B. At the end of 2014, 45 of the stable Baa ratings were withdrawn and the five downgraded credits defaulted. The single-B ratings issued on January 1, 2013 experienced no rating changes, defaults, or withdrawals in 2013. However, in 2014, 10 of them defaulted and 35 were withdrawn.

Based on cohort rating, the average first-year default hazard rate for the Baa category is 0% since no defaults were observed on Baa-rated credits in 2013 or 2014. The average second-year default hazard rate for Baa is 5 / (100 - 45) = 9.09%. (This statistic is based solely on the performance in 2014 of the 100 Baa-rated credits issued on January 1, 2013). Hence, the average two-year CDR for the Baa category is 9.09%.

By original rating, the first-year default hazard rate for the Baa category is 0% since no defaults were observed on Baa-rated credits within the first year of being rated. The second-year default hazard rate considers the second-year performance of the 100 initially-rated Baa credits and is 5 / (100 – 45) = 9.09%. Hence, the two-year cumulative default rate for the Baa category is 9.09%, the same as the two-year cumulative default rate calculated by cohort rating. In the single-B category, however, there is a significant difference.

For the single-B category, the average first-year default hazard rate by cohort rating is (0 + 5 + 10) / (100 + 5 + 100 - 35) = 8.82%. Note both the numerator and denominator include the five single-B credits which were initially rated Baa on January 1, 2013. The second-year default hazard rate by cohort rating is 10 / (100 - 35) = 15.38%. Therefore, the average two-year cumulative default rate is 1 - (1 - 8.82%)*(1 - 15.38%) = 22.84%.

However, by original rating, the first-year single-B default hazard rate is 0% because there are no defaults in 2013 on the 100 single-B credits issued on January 1, 2013. The second-year default hazard rate is 15.38%, which is the same rate calculated by cohort rating. Thus, the single-B two-year cumulative default rate by original rating is 15.38%, which is substantially lower than 22.85%, the cumulative default rate by cohort rating.

The fundamental difference between the single-B two-year default rates by original rating and cohort rating is due to the treatment of the five credits initially rated Baa on January 1, 2013, but downgraded to single-B by January 1, 2014. The performance of these five credits in 2014 enters into the calculation of the first-year default hazard rate by cohort rating, but not by original rating. If the downgraded single-B's default at a higher rate in 2014 than the initially-rated single-B's do in 2013, the default rate calculated by cohort-rating will be higher than the default rate calculated by original rating. Conversely, if the performance of these downgraded single-B's is better, the default rate by cohort rating will be lower instead.

Periodicity

In our definitions of hazard rates and cumulative default rates, we discussed "period *t*" and "period *t+1*" without specifying their durations. While the examples presented above assumed an annual periodicity (meaning default hazard rates are calculated on a yearly basis), Moody's in fact typically calculates default hazard rates at a monthly periodicity. The choice of period will impact the results; however nothing in the above formulations changes whether calculated at an annual, quarterly, monthly, or even daily period.

Loss Rates and Loss Given Default Metrics

Credit Loss Rates

Moody's credit ratings are opinions of relative expected credit loss, which is a function of both the probability of default (PD) as well as the severity of default (the loss given default or LGD). The expected credit loss rate for time horizon T is calculated as the product of the T-horizon average cumulative default rate and the T-horizon average loss severity rate (which can be expressed as one minus the recovery rate):[6]

$$L_T = PD_T * LGD_T = PD_T * (1 - R_T) \qquad (6)$$

[6] We present an example calculation for expected loss below.

Issue-Weighted and Issuer-Weighted Recovery Rates

The table below defines the various concepts of recovery rates which Moody's calculates.[7]

Statistic	Definition
Issue Recovery Rate	The present value of the amount bondholders receive at default resolution divided by the principal amount outstanding as of the default date.
Issuer Recovery Rate	The face value-weighted mean across the recoveries of each individual issue for a particular defaulted issuer. The face value of a bond is the amount paid to the bondholder at maturity.
Issuer-Weighted Mean Recovery Rate	The mean issuer recovery rate. Answers the question, if we select an issuer at random, what is the expected recovery rate?
Value-Weighted Mean Recovery Rate	The face value-weighted mean issuer recovery rate. Answers the question, if we select a defaulted dollar at random, what is the expected recovery rate?
Issuer-Weighted Median Recovery Rate	The median issuer recovery rate. Answers the question, what is the issuer recovery rate around which half of all issuer recovery rates are below and half are above?
Issue-Weighted Mean Recovery Rate	The mean issue (as opposed to issuer) recovery rate . While this measure is widely reported, it is useful only for predicting the expected recovery rate on a portfolio of bonds diversified across issues, ignoring issuer or issue size.

Loss Given Default Rates and Cumulative LGD Rates

The LGD rate is determined using various methods. One can calculate the LGD rate by discounting cumulative payment losses, or alternatively estimate it as one minus the trading price of the defaulted security 30 days after default, or use yet some other estimation technique. When using payment loss discounting, the LGD rate is computed as the sum of the present value of net losses, including both interest shortfalls and principal losses, discounted by the security's coupon rate and expressed as a percentage of the principal balance as of the default date.[8] In other words, the LGD as a percentage of the balance at default $(B_{t_{def}})$ is given by:

$$LGD_{t_{def}} = \frac{\sum_S (IS_S + PL_S) * DF_S}{B_{t_{def}}} \qquad (7)$$

where IS_s and PL_s denote the net interest shortfall and net principal loss in dollars, respectively, at time s and DF_s is the discount factor from the default date to time s.[9]

For non-amortizing securities, we only report $LGD_{t_{def}}$, independent of any time period. For amortizing securities, we typically adjust $LGD_{t_{def}}$ to reflect the balance remaining at time t. To illustrate the details of the adjustment, consider a defaulted credit with balance outstanding from inception date ($t = 1$) to default date ($t = 5$), as shown below. Let B_t and LGD_t denote the balance and LGD rate, respectively, at time t. Assume LGD at default $(LGD_{t_{def}})$ is already known to be 80%. Then, in the periods leading up to default, LGD_t[10] is given by:

$$LGD_t = LGD_{t_{def}} * \frac{B_{t_{def}}}{B_t} \qquad (8)$$

7 This table is derived from Moody's Special Comment, "Recovery Rates on Defaulted Corporate Bonds and Preferred Stocks, 1982-2003," published in December 2003.
8 On occasion, we may also discount net losses by the prevailing market rate.
9 In our research, we commonly refer to $LGD_{t_{def}}$ as "final realized LGD".
10 For simplicity, the conversion of $LGD_{t_{def}}$ to LGD_t does not require any further discounting of losses from the default date to time t.

An Example of Calculating LGD over Time

Time Period (*t*)	Balance (*t*)	LGD (*t*)
1	100	80% * (60 / 100) = 48%
2	90	80% * (60 / 90) = 53%
3	80	80% * (60 / 80) = 60%
4	70	80% * (60 / 70) = 69%
5	60	80%

The following example illustrates how default and LGD rates are combined to calculate the loss rate.

An Example of Calculating a Two-Year Loss Rate

2013			2014		
1/1/2013	12/31/2013		1/1/2014	12/31/2014	
Number Outstanding	Defaulted	Withdrawn	Number Outstanding	Defaulted	Withdrawn
100	5	0	95	6	45
	(3 with LGD=40%, 2 with LGD=15%)			(3 with LGD=60%, 3 with LGD=40%)	

In this example, there are five defaults in the first year: three have an LGD of 40% and two have an LGD of 15%. Note that the LGD rates are as of the cohort date, January 1, 2013. Six credits default in the second year: three have an LGD of 60% and three have an LGD of 40%. Note that these LGD rates are as of the cohort date, January 1, 2014. The average LGD rate for the first year is (3*40% + 2*15%) / 5= 30% and the average LGD rate for the second year is (3*60% + 3*40%) / 6 = 50%.

The one-year cumulative default rate is 5% and the two-year cumulative default rate is 1 - (1 - 5%)*(1 - 6 / (95 - 45)) = 16.4%; therefore, the marginal second year default rate is 16.4% - 5% = 11.4%.[11] Hence, the two-year cumulative loss rate is 5%*30% + 11.4%*50% = 7.2%.

Alternatively, one can calculate the loss rate by first computing the effective LGD rate over the horizon, which is the weighted average of the LGD rates for each year, weighting by the marginal default rate. The cumulative loss rate is then simply the product of the cumulative default rate and the effective LGD rate. In this example the effective two-year LGD rate is (5%*30% + 11.4%*50%) / 16.4% = 43.9%. The two-year cumulative loss rate is 16.4%*43.9% = 7.2%.

[11] The marginal default rate for period *t* is the unconditional probability of a credit defaulting in period *t*. This is distinct from the default hazard rate for period *t*, which measures the probability a credit will default in period *t* conditional on surviving through the beginning of period *t*.

Transition Metrics

A rating transition matrix summarizes the cumulative changes in credit ratings over a given time horizon. It describes the final position of a credit rating given its initial position (though saying nothing about the intervening *path* of ratings). The cells of the matrix are discrete-time estimates of cumulative rating transition probabilities over a particular horizon.[12]

The probability that a credit with rating i when cohort y was formed will transition to rating or state j (which will generally include *default* and *withdrawal*) over a time horizon T is calculated as:

$$p_{ij}(T) = \frac{n_{ij}(y+T)}{n_i\,(y)} \tag{9}$$

where $n_i(y)$ is the number of credits with rating i when cohort y is formed and $n_{ij}(y+T)$ is the number of credits that migrated from rating i to rating or state j, T periods after cohort y is formed.

The weighted average T-period rating transition rate for all cohorts y in the set of cohorts Y is calculated as:

$$\bar{p}_{ij}(T) = \frac{\sum_{y\in Y} n_{ij}(y+T)}{\sum_{y\in Y} n_i\,(y)} \tag{10}$$

Rating transition statistics can be reported by cohort rating or by original rating. For statistics calculated by cohort rating, the cohort includes all credits with outstanding ratings regardless of when the rating was initially assigned. For statistics calculated by original rating, the start date for a particular credit corresponds to the date when it was initially rated and the time horizon corresponds to the age (seasoning) of the rating. For example, in a one-year rating transition matrix by original rating, state i is the original rating and state j is the rating (or default or withdrawal status) held 12 months after the credit was initially rated.

Note that the frequency and magnitude of individual upgrades and downgrades is irrelevant; only the initial rating and the final rating (or, more generally, state) are considered in a transition matrix. Moreover, if a credit is downgraded and then upgraded (or upgraded and then downgraded) so that its initial and final ratings are the same, then for the purposes of a rating transition matrix it will be treated as if its rating never changed. Only net changes are reflected in a transition matrix.

As an example, consider the following portfolio of 100 credits rated B on January 1, 2014. Assume a rating scale with only three possible rating states: A, B and C. The details of a B-rated portfolio at the end of the year are provided below. The details of the A and C portfolios are not provided, although their transition rates are included in the transition matrices for completeness.

[12] Note that since credit ratings are not Markov, transition matrices of different horizons will not generally share an obvious relationship. For instance, squaring the one-year transition matrix will not result in the two-year transition matrix.

Sample Portfolio to Illustrate the Calculation of a Transition Matrix

Portfolio on 1/1/2014, all rated B	100
Portfolio on 12/31/2014	
No rating change or default and still outstanding	60
Upgraded	10
Downgraded to C (includes 5 defaulters, 4 of which were withdrawn)	20
Defaulted but rating was not withdrawn	1
Rating was withdrawn and default occurred subsequently	1
Defaulted and rating was subsequently withdrawn	3
Withdrawn (no associated upgrade, downgrade, or default event)	10

Transition Matrix with Withdrawal and Default States

For this type of matrix, Moody's treats rating changes, rating withdrawals, and defaults as mutually exclusive states, with default and withdrawal as absorbing states. For example, a credit that is downgraded and subsequently defaults within the time horizon is counted as a default, not a downgrade. Similarly, a credit that is upgraded and subsequently has its rating withdrawn within the time horizon is counted as a withdrawal. In cases where a credit both defaults and has its rating withdrawn within the horizon, the event that occurs first determines the final state of the credit with the provision that withdrawn ratings which subsequently default within twelve months, and still within the horizon, are treated as rated defaults.

For the above portfolio the transition matrix with all possible states is shown below.

An Example of a Transition Matrix with Default and WR Columns

	A	B	C	WR	Default
A	80%	10%	1%	7%	2%
B	10%	60%	15%	10%	5%
C	2%	13%	58%	12%	15%

Even though 14% of the B-rated credits were withdrawn by December 31, 2014, only 10% appears in the WR column. In our example, four of the 14 B-rated credits which withdrew ratings also defaulted, and in this case, the defaults occurred within 12 months of the ratings being withdrawn. Hence, those four credits count as defaults rather than withdrawals.

For horizons shorter than or equal to one year, the default rates reported in this type of transition matrix equal the unadjusted default rates described earlier.[13] Consequently, transition default rates are never greater than, and are usually less than, our preferred adjusted default rate calculations.

[13] For horizons longer than one year, it is possible, though unlikely, for a credit to withdraw its rating and default more than one year after the withdrawal, and still within the horizon of study. This credit would be considered a WR in the transition matrix but a defaulter in the unadjusted default rate calculation, resulting in transition default rates that are less than or equal to unadjusted default rates.

Transition Matrix without the Default State

In this type of matrix, the default status of the credit is ignored and only the rating and withdrawal status at the end of the horizon are considered.

An Example of a Transition Matrix with WR Column and No Default Column

	A	B	C	WR
A	80%	10%	1%	9%
B	10%	60%	16%	14%
C	2%	13%	65%	20%

Transition Matrix (without the Default State) using Rating Before Withdrawal

In this type of matrix, the default status of the credit is ignored and the rating prior to withdrawal is used to determine the final position of the credit. Such a matrix has neither a default column nor a WR column, as shown below.

An Example of a Transition Matrix with No Default and WR Columns Using Rating Before WR

	A	B	C
A	87%	10%	3%
B	10%	70%	20%
C	2%	15%	83%

Withdrawal-Adjusted Transition Matrix

Occasionally, a transition matrix is adjusted for rating withdrawals. There are several ways to do this, the simplest being to adjust the denominator for withdrawn credits as follows:

$$\tilde{p}_{ij}(T) = \frac{p_{ij}(T)}{(1-p_{iw}(T))} \tag{11}$$

$$\tilde{p}_{iw}(T) = 0 \tag{12}$$

In effect, credit ratings that are counted as withdrawals in an unadjusted transition rate are removed from the calculation altogether. As Equation (10) shows, withdrawal-adjusted transition rates are equivalent to unadjusted transition rates scaled up by a factor of $\frac{1}{(1-p_{iw}(T))}$. In the example below, the transition rate from B to C is equal to the unadjusted transition rate calculated previously multiplied by a scale factor, or 15% * (1 / (1 – 10%)) = 16.7%.

An Example of a Transition Matrix with Default Column and No WR Column

	A	B	C	Default
A	86%	11%	1%	2%
B	11%	67%	17%	6%
C	2%	15%	66%	17%

Note that this method generally yields higher default rate estimates than the hazard rate method. Hence, Moody's preferred hazard rate-derived adjusted default rate estimates lie between the unadjusted transition default rates and the withdrawal-adjusted transition default rates.

Lifetime Transition Matrix

This type of transition matrix measures the cumulative rating changes of a portfolio of credits starting from the initial rating through a fixed point in time, typically the end of the relevant study period. As mentioned above, transition matrices can be calculated by cohort rating or by original rating. A lifetime transition matrix groups credits by original rating. Rather than study each credit's cumulative transitions over a fixed time horizon, as all other transition matrices we have described thus far do, each credit is studied for a variable horizon defined by the time elapsed from initial rating to the fixed end point. Thus, in this type of transition matrix, it is entirely possible for one credit to be studied for less than one year, and another credit to be studied for a period of decades. Note that a lifetime transition matrix can include any of the configurations of default and WR columns described above.

Volatility Metrics

Volatility metrics measure the incidence, magnitude, and direction of rating changes across a portfolio of credits. Among other applications, they allow us to distinguish portfolios with stable ratings from portfolios with volatile ratings. Additionally, they allow us to perceive trends in the direction of movement of ratings across a portfolio of credits over varying time horizons.

Much like default rates, volatility metrics are calculated for a group of credits over a specified time horizon. Most typically, credits are grouped on the basis of having an outstanding rating as of the cohort formation date, without regard to the specific rating on that date; but it is entirely possible to group credits on the basis of having a common cohort rating or original rating as well.

Note that all volatility metrics can be calculated as both withdrawal-adjusted metrics and unadjusted metrics. For the same reason we prefer adjusted default rates, Moody's prefers to present withdrawal-adjusted volatility metrics. We believe it is more interesting to study the volatility of a group of credits that will remain otherwise outstanding over the specified time horizon. For the majority of volatility metrics that we will discuss, the method of adjustment is identical to that presented for default rates earlier. Thus, for the sake of simplicity, we will focus on the unadjusted definitions for each of the metrics and highlight the withdrawal-adjustment only where it deviates from our standard adjustment methodology.

The following provisions apply for all volatility metrics:

- A credit will only be included if it was outstanding (rated) and not in default as of the cohort formation date.
- If a credit defaults subsequent to the cohort formation date, it is still included in volatility calculations. A defaulted credit can still experience rating changes and these are counted in the metric.
- If a credit's rating is withdrawn after the cohort formation date and subsequently the rating is re-instated, it is no longer a candidate for volatility calculations. In other words, rating changes which occur after the rating is re-instated are not counted in the metric.

In order to understand the various volatility metrics Moody's calculates, consider a portfolio of 20 credits with outstanding ratings on January 1, 2014. The table below shows the path of these 20 ratings over the calendar year, assuming all rating events occur at the end of each quarter. The right half of the table indicates the credit's weight in the corresponding metric's calculation. The metrics themselves are defined subsequently.

Sample Portfolio of Credits to Illustrate the Volatility Metrics Calculations

		Rating in 2014 at the end of:				Weight in the metric									
Credit	Rating on 1/1/2014	Mar	Jun	Sep	Dec	Upgrade Rate	Downgrade Rate	Average Upgraded Notches per Issuer	Average Downgraded Notches per Issuer	Rating Action Rate	Large Upgrade Rate	Large Downgrade Rate	Rating Reversal Rate	Fallen Angel Rate	Rising Star Rate
1	Baa3	Baa3	Ba1	Ba1	Ba1	0	1	0	1	1	0	0	0	1	0
2	Baa1	A3	A3	A3	A3	1	0	1	0	1	0	0	0	0	0
3	Aa1	Aa1	Aa1	A1	A1	0	1	0	3	1	0	1	0	0	0
4	A3	A2	A1	Aa3	Aa3	1	0	3	0	1	1	0	0	0	0
5	Ba1	Baa3	Baa2	Ba1	Ba1	1	1	2	2	1	0	0	1	0	1
6	B1	B2	Caa1	B3	B3	1	1	1	3	1	0	1	1	0	0
7	Caa1	WR	B1	B1	B2	0	0	0	0	0	0	0	0	0	0
8	Caa3	Ca	WR	WR	WR	0	1	0	1	1	0	0	0	0	0
9	B2	B2	B2	B2	B2	0	0	0	0	0	0	0	0	0	0
10	Ba1	Ba1	Ba1	Ba1	Ba1	0	0	0	0	0	0	0	0	0	0
11	B1	B1	B1	B1	B1	0	0	0	0	0	0	0	0	0	0
12	B2	B2	B2	B2	B2	0	0	0	0	0	0	0	0	0	0
13	Ba1	Ba1	Ba1	Ba1	Ba1	0	0	0	0	0	0	0	0	0	0
14	Aaa	Aaa	Aaa	Aaa	Aaa	0	0	0	0	0	0	0	0	0	0
15	Aa2	Aa2	Aa2	Aa2	Aa2	0	0	0	0	0	0	0	0	0	0
16	B1	B1	B1	B1	B1	0	0	0	0	0	0	0	0	0	0
17	B2	B2	B2	B2	B2	0	0	0	0	0	0	0	0	0	0
18	Ba1	Ba1	Ba1	Ba1	Ba1	0	0	0	0	0	0	0	0	0	0
19	Aaa	Aaa	Aaa	Aaa	Aaa	0	0	0	0	0	0	0	0	0	0
20	Aa2	Aa2	Aa2	Aa2	Aa2	0	0	0	0	0	0	0	0	0	0

Upgrade (Downgrade) Rate

The upgrade (downgrade) rate is defined as the number of credits that were upgraded (downgraded) during a specific horizon divided by the total number of credits in the cohort. A credit that was both upgraded and downgraded within the horizon is included in the upgrade and downgrade rate irrespective of its rating at the end of the horizon.

The upgrade (downgrade) rate for a given cohort and horizon answers the question, "for a randomly selected credit, what is the probability it will experience an upgrade (downgrade) at some point within the horizon?" It does not exactly speak to the probability that at the end of the horizon a credit will have a higher (lower) rating.

In the example above, four credits experienced an upgrade in 2014, so the unadjusted upgrade rate is 4 / 20 = 20%.[14] The calculation for the unadjusted downgrade rate is more complicated. Strictly speaking, six credits experienced downgrades in 2014. But Credit #7 experienced a downgrade only after its rating was withdrawn and subsequently re-instated, which disqualifies it from counting in the downgrade rate calculation. Credit #8 also had its rating withdrawn, but because its rating was downgraded prior to the withdrawal, this downgrade counts towards the unadjusted downgrade rate. Overall, five issuers downgraded prior to withdrawal and so the unadjusted downgrade rate is 5 / 20 = 25%.[15]

Note that Credit #5 was upgraded in April and July, but subsequently downgraded in December such that its final rating is the same as its initial rating; it is included in both the upgrade and downgrade rates.

Rating Action Rate

The rating action rate is defined as the number of credits that experienced any rating changes during a specific horizon divided by the total number of credits in the cohort.

The rating action rate for a given cohort and horizon answers the question, "for a randomly selected credit, what is the probability that its rating will change at some point within the horizon?" It does not exactly speak to the probability that a credit's final rating will be different from its initial rating.

In the example above, seven credits in the cohort had a rating action during 2014, so the unadjusted rating action rate is 7 / 20 = 35%.[16, 17] A credit is included in the numerator as long as it had a rating action at any point during the period even though the rating at the start and end may be the same, as for credit #5 above.

Large Upgrade (Downgrade) Rate

The large upgrade (downgrade) rate is defined as the number of credits upgraded (downgraded) by three or more notches over a specific horizon divided by the total number of credits in the cohort. The three-notch change could be a result of a single or multiple rating actions. A credit is considered to have experienced a large upgrade (downgrade) within the given horizon if it crossed the three-notch upgrade (downgrade) threshold relative to its initial rating at any point during the horizon, irrespective of its rating at the end of the horizon.

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14 The adjusted upgrade rate is 4 / (20 – 2) = 22.2%. Credits #7 and #8 withdrew their ratings during the horizon, and consequently are censored from the calculation.

15 The adjusted downgrade rate is 5 / (20 - 1) = 26.3%. Only Credit #7 is censored from the calculation. Even though its rating was withdrawn within the horizon, credit #8 is included in the calculation because its rating was downgraded prior to the withdrawal.

16 Note that Credit #7 experienced a rating action at the end of December, but because it withdrew its rating in April, it does not count towards the rating action rate.

17 The adjusted rating action rate is 7 / (20 - 1) = 36.8%. Credit #7 is censored from the calculation, but not Credit #8.

The large upgrade (downgrade) rate for a given cohort and horizon answers the question, "for a randomly selected credit, what is the probability its rating will rise (fall) three or more notches from its initial rating at some point within the horizon?" It does not exactly speak to the probability that a credit's final rating will be three or more notches different from its initial rating.

Credit #6 in our portfolio crossed the three-notch downgrade barrier in July. It is counted as a large downgrade even though its final rating at the end of the horizon was only two notches below its initial rating. For this portfolio, the unadjusted large upgrade rate is 1 / 20 = 5%; the unadjusted large downgrade rate is 2 / 20 = 10%.[18]

Average Upgraded (Downgraded) Notches per Issuer

The average upgraded (downgraded) notches per issuer is defined as the total number of notches upgraded (downgraded) over a specific horizon divided by the total number of credits in the cohort. For example, a credit downgraded from Ba1 to B1 over 12 months counts with weight three in the average downgraded notches per issuer but only 1 in the downgrade rate.

The average upgraded (downgraded) notches per issuer for a given cohort and horizon answers the question, "for a randomly selected credit, what is the average number of notches it will be upgraded (downgraded) over the length of the horizon?" Note that the average upgraded (downgraded) notches per issuer estimates the unconditional number of notches a credit is expected to be upgraded (downgraded) during the horizon.

Consider the unadjusted average upgraded notches per issuer for our example portfolio above. Credit #4 was upgraded three notches and credit #5 was upgraded two notches. Credits #2 and #6 were upgraded one notch. The unadjusted average upgraded notches per issuer is therefore (1 + 3 + 2 + 1) / 20 = 0.35. Similarly, the unadjusted average downgraded notches per issuer is (1 + 3 + 2 + 3 + 1) / 20 = 0.50.

The adjusted metrics are calculated in the same way as the unadjusted metrics, with the exception that withdrawn credits are immediately censored from the calculation. The adjusted average upgraded notches per issuer is (1 + 3 + 2 + 1) / 18 = 0.39. The adjusted average downgraded notches per issuer is (1 + 3 + 2 + 3 + 1) / 19 = 0.53.

To calculate the multi-period adjusted cumulative average upgraded (downgraded) notches per issuer, we use a different formula than what we presented for adjusted cumulative default rates previously. We define the number of credits outstanding as of the start of period *t* as:

$$n_t = n_{t-1} - w_t \tag{13}$$

where, just as before, w_t is the number of credits which withdraw during period *t*.[19] Let v_t be the total number of notches upgraded (downgraded) in period *t* across all credits outstanding as of the beginning of period *t*. Then the t^{th}-period contribution to the cumulative average upgraded (downgraded) notches per issuer, u_t, is given by:

$$u_t = \frac{v_t}{n_t} \tag{14}$$

[18] The adjusted large upgrade rate is 1 / (20 - 2) = 5.6%. Credits #7 and #8 are censored from the calculation. The adjusted large downgrade rate is 2 / (20 – 2) = 11.1%. Again, credits #7 and #8 are censored from the calculation.

[19] Note that credits which upgraded or downgraded prior to the current period are not censored from the sample. By necessity, they must remain in the sample because they may yet experience rating actions which should also be counted in the metric.

and the cumulative average upgraded (downgraded) notches per issuer, V_t is given by:

$$V_t = \sum_{s=1}^{t} u_s \tag{15}$$

Average Notches per Upgrader (Downgrader)

The average notches per upgrader (downgrader) is defined as the average upgraded (downgraded) notches per issuer divided by the upgrade (downgrade) rate.[20]

The average notches per upgrader (downgrader) for a given cohort and a given horizon answers the question, "given that a credit will be upgraded (downgraded) during the horizon, what is the expected total number of notches it will change across all its upgrade (downgrade) events during the horizon?" Note that the average notches per upgrader (downgrader) estimates the number of notches a credit is expected to be upgraded (downgraded) conditional on a credit having upgraded (downgraded).

For our sample portfolio, the unadjusted average notches per upgrader is 0.35 / 0.20 = 1.75. The unadjusted average notches per downgrader is 0.50 / 0.25 = 2.00.[21]

Rating Drift

Rating drift is defined as the average upgraded notches per issuer minus the average downgraded notches per issuer. It measures the net average number of notches a credit will change over the length of the study horizon.[22]

For our sample portfolio, the unadjusted rating drift is 0.35 – 0.50 = -0.15.[23]

Rating Volatility

Rating volatility is defined as the average upgraded notches per issuer plus the average downgraded notches per issuer. It measures the gross average number of notches a credit will change over the length of the study horizon.[24]

For our sample portfolio, the unadjusted rating volatility is 0.35 + 0.50 = 0.85.[25]

Rising Star Rate

Rising stars are credits whose ratings move from Ba1 or lower (speculative-grade) to Baa3 or higher (investment-grade) over a specified study horizon. A credit is considered a rising star as long it crosses this barrier during the horizon irrespective of its rating at the end of horizon. The rising star rate is the number of rising stars divided by the number of speculative–grade credits in the cohort.

20 The unadjusted average notches per upgrader (downgrader) is equal to the unadjusted average upgraded (downgraded) notches per issuer divided by the unadjusted upgrade (downgrade) rate. Similarly, the adjusted average notches per upgrader (downgrader) is equal to the adjusted average upgraded (downgraded) notches per issuer divided by the adjusted upgrade (downgrade) rate. For a single cohort over a single period, the unadjusted average notches per upgrader (downgrader) is equal to the adjusted average notches per upgrader (downgrader). The same cannot be said for the multi-period average notches per upgrader (downgrader).

21 The adjusted average notches per upgrader is 0.389 / 0.222 = 1.75 and the adjusted average notches per downgrader is 0.526 / 0.263 = 2.00.

22 The unadjusted rating drift is equal to the unadjusted average upgraded notches per issuer minus the unadjusted average downgraded notches per issuer. Similarly, the adjusted rating drift is equal to the adjusted average upgraded notches per issuer minus the adjusted average downgraded notches per issuer.

23 The adjusted rating drift is 0.389 – 0.526 = -0.14.

24 The unadjusted rating volatility is equal to the unadjusted average upgraded notches per issuer plus the unadjusted average downgraded notches per issuer. Similarly, the adjusted rating volatility is equal to the adjusted average upgraded notches per issuer plus the adjusted average downgraded notches per issuer.

25 The adjusted rating volatility is 0.389 + 0.526 = 0.92.

The rising star rate answers the question, "what is the probability that a randomly selected speculative-grade credit will become investment-grade at any point within the study horizon?" It does not exactly speak to the probability that a speculative-grade credit will be investment-grade at the end of the study horizon.

There are 12 speculative-grade credits in our portfolio as of January 1, 2014. Credit #5 is considered a rising star because it was upgraded to Baa3 in April even though its final rating was still speculative-grade. Thus the unadjusted rising star rate is 1 / 12 = 8.3%.[26]

Fallen Angel Rate

Fallen angels are credits whose ratings move from Baa3 or higher (investment-grade) to Ba1 or lower (speculative-grade) over a specified study horizon. A credit is considered a fallen angel as long it crosses this barrier during the horizon irrespective of its rating at the end of horizon. The fallen angel rate is the number of fallen angels divided by the number of investment–grade credits in the cohort.

The fallen angel rate answers the question, "what is the probability that a randomly selected investment-grade credit will become speculative-grade at any point within the study horizon?" It does not exactly speak to the probability that an investment-grade credit will be speculative-grade at the end of the study horizon.

There are eight investment-grade credits in our portfolio as of January 1, 2014. Credit #1 is considered a fallen angel because it was downgraded to Ba1 in July. Thus the unadjusted fallen angel rate is 1 / 8 = 12.5%.[27] Note that Credit #5 is not considered a fallen angel even though it experienced a downgrade from investment-grade to speculative-grade. This is due to the fact it was rated speculative-grade as of the cohort formation date, and hence not eligible to enter the calculation.

Rating Reversal Rate

Rating reversals are defined with respect to a look-back horizon as well as the usual look-forward horizon. A rating action which occurs during the look-forward horizon is classified as a reversal if it is in the opposite direction of the last observed rating action which occurred within the look-back horizon from that date. The rating reversal rate is the number of credits which experience a rating reversal divided by the number of credits in the cohort.

The rating reversal rate answers the question, "what is the probability that a randomly selected credit will experience a rating action reversal (defined with respect to the look-back horizon) at any point during the (look-forward) study horizon?"

For our sample portfolio, the unadjusted rating reversal rate is 2 / 20 = 10%.[28] The two cases in this portfolio are obvious reversals since both the original rating action and the reversal occurred in 2014 (credits #5 and #6).

A more typical reversal would be as follows. Suppose a credit was upgraded on June 30, 2014. Whether that change constitutes a "reversal" depends on whether a downgrade occurred within the look-back horizon from June 30, 2014. If the look-back horizon was 12 months (as it is typically), and if the credit had been downgraded on September 20, 2013, then the June 30, 2014 upgrade would be classified as a "reversal."

[26] The adjusted rising star rate is 1 / (12- 2) = 10%. Credits #7 and #8 are censored from the calculation.

[27] The adjusted fallen angel rate is also 1 / 8 = 12.5%. Even though Credits #7 and #8 withdrew, they do not affect this calculation because they were not rated investment-grade as of the cohort formation date.

[28] The adjusted reversal rate is 2 / (20 – 2) = 11.1%. Credits #7 and #8 are censored from the calculation.

Accuracy Metrics

Accuracy metrics measure the relative or ordinal accuracy of ratings. In addition to assigning low ratings to credits that ultimately default, the effectiveness of a rating system depends on its ability to assign high credit ratings to credits that do not default. In other words, an efficient rating system is able to distinguish low default-risk credits from high default-risk credits. The ratings accuracy metrics described below are useful in measuring the relative rank-ordering of credit risk under a given rating system.

These metrics can be used to measure (1) the performance of a single cohort at a point in time, (2) the average performance of a sequence of cohorts over time, or (3) the performance of a pool of ratings drawn from different cohorts over time.

Cumulative Accuracy Profile (CAP) Plot

The CAP plot measures the cumulative share of defaults accounted for by the cumulative share of ratings.[29] Specifically, the share of credits rated equal to or lower than a given rating is measured along the *x*-axis while the share of defaults accounted for by those credits is measured along the *y*-axis. The CAP curve plots what percentage of defaults *y* is accounted for by the first *x* percent of ratings. Since all defaults must be accounted for within all the ratings, $y = 1$ when $x = 1$, but ideally $y = 1$ before $x = 1$.

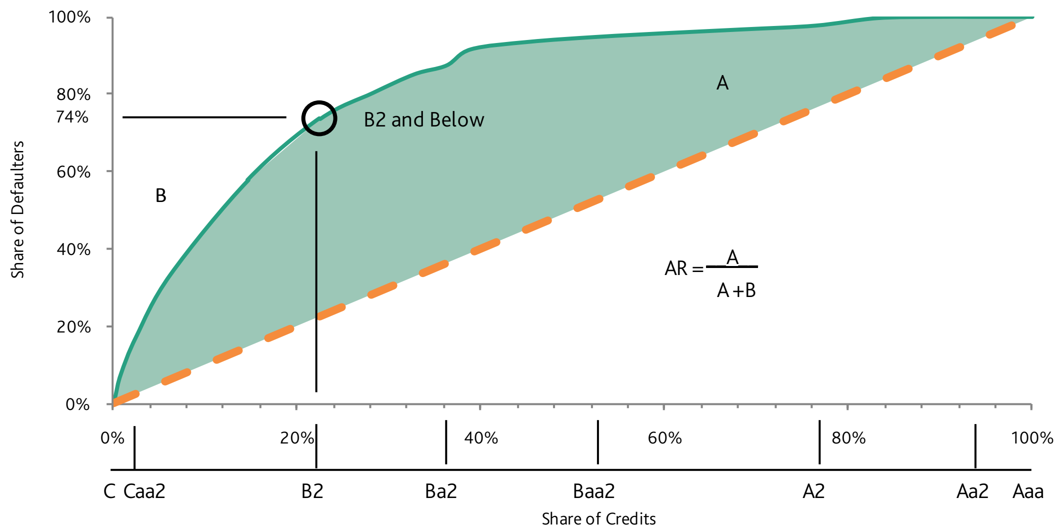
If a rating system had no correlation to defaults, meaning defaults were uniformly distributed through the rating scale, then we would expect that the first x% of ratings (measured from the bottom) would contain about x% of the defaults. The CAP curve would fall along the 45°-line, y = x. To the extent the rating system does offer some discriminatory power, then among the first x% of ratings we should find y% > x% of the defaults – in other words, the CAP curve should bow out toward the upper left corner (y > x). In a perfect system, all of the lowest-rated credits would default and all defaulters would hold the lowest rating, meaning y = 1 everywhere.

Each point along a CAP curve indicates the power of a specific rating category as a tool to discriminate between defaulters and non-defaulters over a specified investment horizon, such as one year, five years, or twenty years. A single-cohort CAP curve evaluates the performance of a group of ratings that were outstanding at the same specific point in time, whereas a multiple cohort CAP curve evaluates a pool of ratings that were outstanding at different points in time.

As an example, refer to the CAP plot below based on the January 1, 2012 cohort of corporate ratings and the defaults which occurred during the subsequent three years. The percentage of defaulters during 2012 – 2014 captured by each rating category and below is plotted along the vertical axis, while the share of credits with ratings at or below each rating category is plotted along the horizontal axis. For example, credits rated B2 or below represent 22% of the overall population but account for 74% of the defaulting population, so the plot contains the point (0.22, 0.74).

[29] For a more detailed discussion of ratings accuracy metrics, please refer to Moody's Special Comment, "Measuring The Performance Of Corporate Bond Ratings," published in April 2003.

EXHIBIT 1
Cumulative Accuracy Profile (CAP) Plot and Accuracy Ratio Calculation



Accuracy Ratio

The accuracy ratio (AR) compresses the information contained in the cumulative accuracy profile into a single statistic. The accuracy ratio is the ratio of the area between the CAP curve and the 45° line (area A in the figure above) to the total area above the 45°-line (areas A+B in the figure below). Note that the AR is not a sufficient statistic of the CAP curve: while a given CAP curve implies only one AR, multiple CAP curves could imply the same AR.

A discrete rating system, even the optimal system, cannot obtain an AR = 1. We therefore typically adjust the AR to compensate for this so that in principle it is possible to achieve an AR = 1 (and, for that matter, an AR = -1). Let *D* denote the (unadjusted) default rate. Then from the unadjusted *AR* we obtain the adjusted *AR** as follows:

$$AR^* = \frac{AR}{1-D} \qquad (16)$$

Average Defaulter Position

The Average Position of defaults[30] (AP) also measures the ordinal power of rating and is Moody's preferred metric for measuring ratings accuracy. It is equivalent to the AR but offers a more straightforward interpretation. Define the *position* of any credit in the cohort as the share of credits rated better than it, assuming each credit occupies the midpoint of its rating category. For example, the position of every Aa2 credit is the share of the cohort rated Aaa or Aa1 plus half the share rated Aa2. The AP is simply the average position of the defaulted credits. Intuitively, a more powerful rating system should have low rated defaults and high rated non-defaulters, meaning the average position (as we have defined it) of defaulters should be high.

[30] For a detailed discussion of average defaulter position and mathematical derivation of accuracy ratio from average defaulter position, please refer to Moody's Special Comment, "Measuring Ratings Accuracy Using the Average Default Position," published in February 2011.

Consider a rating system with five rating categories and 1,303 ratings outstanding on January 1, 2012 shown in the table below. Over a three-year horizon 128 of these were in default.

An Example of Calculating Average Defaulter Position

Rating	Credits	Defaults	Position
Aaa	157	0	6.0%
Aa	198	1	19.6%
A	225	7	35.9%
Baa	355	36	58.1%
Ba	268	29	82.0%
C	100	55	96.2%
Total	1303	128	
Average Defaulter Position			78.4%
Adjusted Average Defaulter Position			86.9%

For credits rated Aaa, there are of course no credits with higher ratings. Still, there is some ambiguity as to where within the Aaa band each credit resides. The Aaa band is exactly 157 / 1,303 = 12% wide. We choose to place all credits at the midpoint of the Aaa band and thus calculate the position of each Aaa credit as 6%. The midpoint of A is (157 + 198 + 225 / 2) / 1303 = 35.9%.

Once the positions of each rating category are established, the average defaulter position AP is simply the average position of all the defaulters in the cohort. There is one Aa rated defaulter with a position of 19.6%, seven A-rated defaulters with positions of 35.9%, and so on down to 55 C-rated defaulters with positions of 96.2%. Thus AP = (6.0%*0 + 19.6%*1 + 35.9%*7 + 58.1%*36 + 82.0%*29 + 96.2%*55) / 128 = 78.4%.

AP is bounded between 0% and 100%, with 100% indicating perfect sorting power, 50% indicating no power, and 0% indicating perfectly negative power. But as is the case with AR, a discrete rating system can never obtain an AP of 100% (or 0%), hence we make an adjustment. Once again letting D denote the unadjusted default rate for the cohort, we obtain the adjusted AP* as follows:

$$AP^* = \frac{AP - \frac{1}{2}}{1 - D} + \frac{1}{2} \quad (17)$$

For our example, D = 128 / 1303 and the adjusted AP is 81.5%.

It should be noted that Moody's routinely reports adjusted average positions which we denote simply AP, not AP*. Should we ever report an unadjusted average position, we would label that "unadjusted AP."

The AP conveys the same information as the AR. In particular, whether comparing adjusted metrics or unadjusted metrics, it can be shown that:

$$AR = 2 * AP - 1 \quad (18)$$

For the cohort above, AR = 2*0.784 - 1 = 57% and the adjusted AR is 2 * 0.815 -1 = 63%.

In addition, just as the AR is derived from the CAP curve, it can be shown that the unadjusted AP is simply the total area under the CAP curve, including the area under the 45° line.

Adjusting for Withdrawals

The AP, AR, and CAP curve can also be calculated by removing credits whose ratings have been withdrawn before the end of the time horizon. Removing withdrawals treats them as (immediately) censored observations while including them implicitly treats them as non-defaulters.

Adjusting for Outlook and Watchlist Assignment

While ratings are primary indicators of credit quality, Outlooks and Watchlist assignments are supplemental indicators used by Moody's to communicate potential changes in credit quality. A Moody's rating Outlook is an opinion regarding the likely direction of the credit quality of an issuer or issue, and therefore its rating, over the medium term. Outlooks may be positive, negative, or stable and, very rarely, "developing" meaning the Outlook itself is contingent upon an unresolved event.

Rating reviews, or Watchlist assignments, are stronger statements about expected near-term rating actions. Issuers or issues may be placed on Watch for an upgrade or downgrade or, again rarely, for an "uncertain" direction. Watch assignments supersede Outlooks.

Moody's research indicates that Outlook and Watch assignments can contribute important supplemental information about relative default risk over and above the credit rating alone.[31] Adjusting ratings for Outlook and Watchlist assignments leads to outlook adjusted Average Position or Accuracy Ratio metrics. Specifically, before calculating AP or AR we first adjust credit ratings as follows:

» Lower (raise) ratings by one notch for negative (positive) Outlooks; and

» Lower (raise) ratings by two notches for credits on Watch for downgrade (upgrade).

» Constrain the resulting adjusted ratings to be Aaa or lower and single-C or better.

Once the cohort ratings are adjusted, the AP or AR is calculated in the usual manner. For example, a Aa3 credit with a positive Outlook is considered to be a Aa2 credit for purposes of calculating the AP or AR. Similarly, a B1 credit on Watch for a downgrade is treated as a B3 credit.

Adjusting for LGD

The AP or AR, as well as the CAP curve, may be adjusted to reflect the loss given default (LGD) of credits. To calculate the LGD-adjusted AP, defaulted credits are given weight equal to their LGD. If all defaults lead to total loss (really, if all defaults have identical non-zero loss), then the LGD-adjusted AP would equal the simple AP. The LGD-adjusted CAP curve would report the share of loss – rather than the share of default – as a function of the share of ratings.

[31] For details, please refer to Moody's Special Comment, "Effects of Watches, Outlooks, and Previous Rating Actions on Rating Transitions and Default Rates," published in August 2011.

Frequently Used Terms

Default and Impairment

Please refer to the section on "Definitions of Default, Impairment and Loss-Given-Default" in Moody's latest Rating Symbols and Definitions document, which can be accessed on the Ratings Definitions page on moodys.com. To the extent a particular study from Moody's employs a modified definition of default or impairment, that will be noted within the report itself.

Median Rating Prior to Default (Adjusted for LGD)

For credits which defaulted within a fixed interval of time, the median rating before default is defined as the median of these credits' ratings at a specified period of time before default. Typically, we consider defaults within a one year interval and take the median of their ratings 12 months prior to the default date.

The median rating prior to default may be adjusted for LGD by first weighting each defaulted credit by its estimated final LGD.

Broad Ratings and Refined or Alphanumeric Ratings

Broad ratings refer to the following Moody's long-term bond rating categories: Aaa, Aa, A, Baa, Ba, B, and Caa or below. Refined ratings, sometimes called specific or alpha-numeric ratings, include numeric modifiers: Aaa, Aa1, Aa2, Aa3, A1, A2, A3, Baa1, Baa2, Baa3, Ba1, Ba2, Ba3, B1, B2, B3,Caa1, Caa2, Caa3, Ca, and C.[32]

Investment-Grade (IG) and Below Investment-Grade (BIG)/Speculative-Grade (SG) Ratings

Investment-grade ratings refer to Aaa, Aa1, Aa2, Aa3, A1, A2, A3, Baa1, Baa2, and Baa3. Below investment-grade or speculative-grade ratings refer to Ba1, Ba2, Ba3, B1, B2, B3, Caa1, Caa2, Caa3, Ca, and C.[10]

Investment-Grade Default

We classify a default as an investment-grade default if the credit held an investment-grade rating at any point within a specified period of time (typically 12 months, unless otherwise stated) prior to default.

Share of Credits on Review for Upgrade (Downgrade)

The share of credits on review for upgrade (downgrade) is defined as the number of credits on review for upgrade (downgrade) as of a given date divided by the total number of credits outstanding as of that date.

Watch / Outlook Index

The watch/outlook index is a summary of the watch and outlook designations for the credits in a particular cohort. The metric is calculated by first assigning the following weights to each credit: +2 if on review for upgrade, +1 for a positive outlook, -1 for a negative outlook, and -2 if on review for downgrade. The watch/outlook index is computed as the sum of these weights divided by the total number of credits. This metric provides a summary of Moody's view of the likely direction and severity of ratings changes in a cohort in the short to medium term.

Cohort Count

The cohort count is defined with respect to any ratings performance statistic which averages over multiple cohorts. The cohort count is equal to the total count of cohorts which support the average statistic. We commonly include this metric alongside an average statistic (e.g., average cumulative default for Baa-rated credits from 2005 – 2015, one-year average defaulters' position over the time period 1983 – 2015) to

[32] For structured finance securities, these ratings may include the (sf) indicator.

indicate the robustness of the average. Averages which are supported by a large cohort count are more robust than those supported by smaller cohort counts.

Average Cohort Size

The average cohort size is a companion metric to the cohort count. It is defined as the average cohort size for all cohorts which support a particular average ratings performance statistic. Larger average cohort sizes once again are indicative of a more robust statistic.

Report Number: 182319

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